

The role of pragmatic markers in perceptions of L2 fluency in dialogue

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ABSTRACT

This matched-guise study helps identify the features of interaction that contribute to perceptions of fluency in L2 dialogue. We focus on pragmatic markers (PM) that occur frequently in L1 interaction but relatively less so in L2 English learners' speech (Fung & Carter, 2007), namely, interpersonal (e.g., *I see, okay*) and cognitive PMs (e.g., *I mean, well, I think*). Despite evidence of their functional value in interaction (McCarthy, 2009), they are often considered markers of disfluency and rarely introduced in L2 classrooms. Speech samples of highly proficient L2 interaction were digitally manipulated to control for interpersonal and cognitive PMs simulating expert speaker-like PM use. L1 and L2 English listeners ($N = 231$) rated and commented upon interlocutor's fluency. The quantitative results suggest that listeners generally perceived speakers as more fluent when using PMs, although the magnitude of the effect was relatively small and driven by L1 raters. Raters' qualitative comments suggest covert influence of PM use on fluency perceptions and uncovered a bias favoring English L1 speaker PM use. Results support the potential value of explicitly addressing PMs in teaching L2 fluency and help researchers in understanding the construct of spoken fluency in dialogue to guide in its measurement and assessment.

1. Introduction

Spoken fluency often constitutes one of the main goals for L2 learners given the negative consequences (e.g., losing face) of perceived disfluency (Lennon, 1990), and the crucial role of being perceived as orally fluent during high stakes oral proficiency tests, such as the TOEFL or the OPI. From a pedagogical perspective, there are clear advantages for both L2 learners and language teachers to develop empirically-based understanding of the language features that have a positive impact on judgements of perceived fluency (Tavakoli et al., 2016; Tavakoli & Hunter, 2018). However, from a theoretical standpoint, much of our understanding of what influences listeners' perception of L2 fluency has been limited to the study of monologic speech and observable temporal features of fluency (Segalowitz, 2016), in terms of breakdown, repair, and speed (Tavakoli, 2016). While speed and pausing phenomena (frequency, duration, and placement) have a clear impact on listener-based judgments of fluency (Suzuki et al., 2021), features of speech characteristic of interactional competence and the co-construction of fluent communication have generally been overlooked in experimental research. Building on the work of Van Os et al. (2020), which found turn-taking behavior to influence perceptions of L2

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dialogue fluency, this study aims to provide a more nuanced understanding of what listeners perceive as fluent L2 interaction. We shift focus to another set of language features identified as contributing to productive and perceived oral fluency associated with dialogic speech—pragmatic markers (henceforth, PMs) (e.g., Götz, 2013; Hasselgren, 2002; Wolk et al., 2021).

PMs encompass a wide range of language features (e.g., modal particles, general extenders, routines) that have been associated with sounding “native-like” or “idiomatic” (Wolk et al., 2021) and, more importantly, with higher perceived fluency (e.g., Hasselgren, 2002). However, corpus research has revealed that they are used differently by L2 learners of English compared to English L1 speakers in terms of frequency, type, placement, and variety of use (Fung & Carter, 2007), with a preference for alternative fluency enhancing strategies, such as filled pauses, “which render the speech less fluent” (Wolk et al., 2021, p. 29). Motivated by these findings, we aim to better understand the potential role of PMs in explaining the nuanced phenomenon of perceived fluency in the successful co-construction of dialogic speech through a broader problem-solving perspective (Peltonen, 2017), in which speakers utilize communication strategies as *fluency resources* in coping with processing time pressure and lexis-related obstacles (p. 1–2). The role of PMs in shaping our perceptions of fluency is particularly important as language learners are often confronted with assessments of their oral performance based, at least to some extent, on raters’ intuition about what it means to be fluent (e.g., “*You rely too much on fillers*”). Our review of the literature begins by contextualizing the construct of fluency in L2 research. We highlight discrepancies in its conceptualization and a recent shift in attention from monologic to dialogic fluency. We then explore the work that has considered the influence of PMs on perceived fluency in interactive speech.

1.1. Perceived fluency

Impressionistic judgments of oral fluency represent an important layer in developing a complex definition of fluency because in practice, fluency is often operationalized as a listener construct, particularly when raters judge fluency as a distinct component on performance assessments (e.g., oral proficiency exams). Subjective and objective measures are necessary to understand fluency among second/foreign language users (L2 fluency), particularly since the notion of fluency, in its broad sense, is so deeply embedded in lay perceptions of L2 speech and in academic and professional considerations.

Prior research has yielded mixed results regarding how L1 and L2 raters judge L2 spoken performance, and L2 fluency in particular (Kormos & Dénes, 2004). It has been suggested that L1 speakers might rate L2 speakers’ oral proficiency lower due to inherent biases, particularly in English home language countries (e.g., Nguyen, 1993), making L2 speakers susceptible to negative linguistic stereotyping. Likewise, it has also been indicated that L2 speakers might rate other L2 speakers’ language production either harsher or with more leniency than L1 speakers depending on their respective language backgrounds and their similarities and/or familiarity with the L2 speakers they are rating (Bent & Bradlow, 2003). Using a mixed-methods design, Magne et al. (2019) found that L2 raters’ fluency judgements were associated with temporal features as well as perceptions of accent, pronunciation, and L1-L2 status.

To be sure, untangling the construct of perceived fluency is complicated, particularly as perceptions are likely also tied to contextual variables such as topic, setting, planning time, and interlocutor. Researchers have taken different approaches in measuring perceptions of fluency—some provided no specific instructions to raters (e.g., Kormos & Dénes, 2004), whereas others, such as Derwing et al. (2004) and Rossiter (2009) specifically instructed raters to attend to temporal features (fluency in the narrow sense), such as filled and unfilled pauses. Several objective measures of temporal speech have been found to predict perceptions of fluency, such as number of unfilled pauses per second, location of unfilled pauses, speech rate, and pace (Derwing et al., 2004; Kormos & Dénes, 2004; Iwashita et al., 2008; Rossiter, 2009). Other linguistic features that have been hypothesized to influence raters’ perceptions of fluency, although with less empirical support, include grammatical accuracy, lexical diversity (Iwashita et al., 2008), and pronunciation or accentedness (Kormos & Dénes, 2004; Rossiter, 2009).

Research investigating perceptions of spoken fluency has traditionally been limited to the examination of monologic rather than dialogic speech due to practicality and control, but this limitation represents a significant gap in the development of a complete understanding of fluency as a construct (McCarthy, 2009; Suzuki et al., 2021; Tavakoli, 2016). A growing number of studies have sought to identify the features that differentiate fluent production between monologic and dialogic speech. For example, Tavakoli (2016) and Witton-Davies (2013) found that dialogic performance resulted in faster speech rates, less pausing, and fewer repairs than in monologues. They speculated that increased listening and planning time contribute to ease in conceptualization and delivery. In a recent experiment, Van Os et al. (2020) acoustically manipulated speech in dialogic performances to find hesitation in answering questions to negatively influence perceptions of L2 fluency. Tavakoli (2016), along with Van Os et al. (2020), argued that while monologic fluency has been predominantly characterized by speed, flow, and lack of disfluency measures, in dialogue, speakers’ conversational and accommodational skills gain priority. This argument is crucial to the theoretical underpinnings of the current study. Rather than an exclusive focus on speed and flow, dialogue fluency may be better demonstrated in interaction by incorporating skill in turn-taking (Van Os et al., 2020), negotiation of meaning, appropriate overlap, and other collaborative features of interactional competence, such as the use of PMs, to which we turn next.

1.2. The role of pragmatic markers in perceived spoken fluency

As particles that do not always add to the propositional meaning of an utterance, for a long time, PMs were, to some extent, looked down on as markers of “unclear thinking, lack of confidence, [or] inadequate social skills” (Crystal, 1988, p. 47Crystal, 1988) and at times discounted as “disfluency phenomena” (Trenchs-Parera, 2009, p. 373). Many of them are to this day considered bad habits, such as the repeated use of *like* or *and stuff*. Yet, research has shown that depending on the context PMs have a wide variety of discursive functions (Aijmer & Simon-Vandenberg, 2011). PMs can be used to “signal transitions in the evolving process of conversation, index

the relation of an utterance to the preceding context, and indicate an interactive relationship between speaker, hearer, and message" (Fung & Carter, 2007, p. 411). Importantly for our purposes, they also play a role in perceptions of oral fluency (e.g., Hasselgren, 2002). From an assessment perspective, "pragmatic competence" has been recognized as a conception of fluency by CEFR, suggesting greater interest in the features of discourse competence (Council of Europe, 2020). Prior fluency research, however, suggests that L2 speakers use fewer PMs than L1 speakers, and prefer the use of other fluency-enhancing strategies instead, such as filled pauses (e.g. *ehm*) (Götz, 2013; Wolk et al., 2021).

In this study, we adopt Fung and Carter's (2007) PM definition and functional taxonomy, within which PMs fall in four functional categories: interpersonal, cognitive, referential, and structural. Interpersonal PMs serve to mark shared knowledge (e.g., *(you) see, you know*), indicate attitudes (e.g., *well, really, I think*), and show responses (e.g., *okay, right/alright, (oh) great*). Cognitive PMs include those with the function of denoting thinking processes (e.g., *well, I think, I see*), reformulation or self-correction (e.g., *I mean, that is*), elaboration (e.g., *like, I mean*), hesitation (e.g., *well, sort of*), and assessment of the listener's knowledge (e.g., *you know*). Referential markers express cause, comparison, contrast, coordination, disjunction, and consequence (e.g., *but, because, and*) while structural markers function to open, close, shift, and continue topics and sequence events (e.g., *now, alright, how about*).

Fung and Carter's corpus-based analysis revealed that the classroom language of L2 learners of English in Hong Kong secondary schools incorporated more restricted use of interpersonal and cognitive markers, relative to L1 speakers' classroom interaction in secondary classrooms in the UK (taken from a sub-corpus of the Cambridge and Nottingham Corpus of Discourse English, or CAN-CODE). In contrast, their L2 learners used referential markers more frequently than their L1 counterparts, although with less variety. Fung and Carter's results, then, align with prior research comparing PM use by language learners and L1 speakers, which has uncovered a tendency for over-generalization in use of some PMs types by L2 learners, e.g., *well* for advanced Swedish learners of English (Aijmer, 2011), restricted use of interpersonal PMs (Wei, 2011), and disparities in diversity and positioning of PMs (Hasselgren, 2002). Their functional taxonomy, however, exposes the potential for a pedagogically fertile difference in patterns of use. That is, learners' more substantial use of referential and structural PMs was not completely unexpected because, as Fung and Carter point out, referential PMs are more likely to be taught than other types of PMs in L2 classrooms, particularly within the context of writing (i.e. discourse organization, cohesiveness, topic organization). However, the reported underuse of interpersonal and cognitive PMs by L2 learners of English, compared to their L1 counterparts, is perhaps indicative of the fact that they are rarely taught (Hellermann & Vergun, 2007; Rossiter et al., 2010), and raises the question of the potential value of including them in L2 curricula.

Existing research suggests the positive effect PMs can have on interactional fluency in terms of automaticity, as speakers can employ chunks to improve flow (e.g. *that sort of thing, on the other hand*), provide continuity with preceding utterances for smooth transitions (turn openers, such as *well* or *yeah*), or trigger speaker-change (turn-closing moves such as *and stuff, I know*) (McCarthy, 2009). The majority of studies that have examined the relationship between PMs and oral proficiency (e.g., Wei, 2011) or fluency (e.g., Götz, 2013; Hasselgren, 2002) have taken a descriptive corpus linguistic approach. Götz (2013) argues that PMs contribute to L1 speakers' productive fluency by (a) "increas[ing] the general degree of naturalness in speech substantially", and (b) by increasing "the degree of temporal fluency" when "used as a speech planning strategy, especially when a lexical gap emerges" (resulting in fewer filled or unfilled pauses) (p. 39). Evidence is clear of the processing advantages of the use of formulaic sequences in reducing length of pauses in monologic elicited imitation tasks (Yan, 2019), and in turn, we predict that some multiword PMs influence listener perceptions in similar fashion. From an interactive perspective, Hasselgren (2002) goes a step further and connects the expert-like use of PMs ('smallwords' in her denomination) with perceived fluency, arguing that PMs constitute a system of 'prototypical linguistic cues' that serve to facilitate the transfer of communicative intention thereby "putting a minimum of strain on the listener" (p. 156). In her corpus-based study comparing PM use between L1 Swedish students of English and L1 speakers, Hasselgren found that students overused particular PMs in contexts in which L1 speakers demonstrated a wider range, and that a wider, more expert-like range of PMs was associated with higher fluency. Mounting descriptive evidence from corpus research that highlights the disparities in production of PMs between L1 speaker (or expert) speech and L2 learner speech compels investigation into whether this divergence manifests in perceptions of the "degree of naturalness of the output" (Götz, 2013, p. 40). However, to the best of our knowledge, no experimental studies thus far have measured the effects of PM use on listeners' perceptions of L2 speakers' oral fluency. To begin to address this gap, the present study investigates the potential influence of PMs that are under-utilized by L2 speakers (interpersonal and cognitive PMs) on perceptions of fluency in L2 dialogue. We focus on the potential influence of L1-L2 listener status because we anticipate that the differences in how L1 speakers and L2 speakers use PMs may influence perceptions (e.g., expectations) of their use, particularly as L2 English speakers increasingly use English as a lingua franca among each other.

1.3. Research questions

To better understand the construct of fluency in interaction, and in turn, the potential functional value of PMs, we ask the following research questions.

RQ1. To what extent does expert-like use of interpersonal and cognitive PMs influence perceived fluency of English L2 interactive speech?

RQ2. Does the L1-L2 identity of raters affect these ratings?

2. Material and methods

2.1. Overview of the research design

The experiment was designed to compare fluency ratings between two segments of the same L2 speakers' interactive speech with the presence/absence of expert-like use of interpersonal and cognitive PMs as the independent variable. The following principles were prioritized in guiding the research design.

1. *Create speech samples from authentic conversations rather than scripted dialogue.* We consider the construct of interactive fluency as dynamic, improvisational, and spontaneously reactive, involving real time adjustment as interlocutors' co-construct fluid exchange, i.e., confluence (McCarthy, 2009). An argument underpinning this study is that cognitive and interpersonal PMs may serve as a subset of tools that are engaged in the construction of interactive fluency. While scripted dialogue would allow for straightforward control of the stimuli, we felt it crucial for the L2 speakers to own their use of PMs to maintain the perception of natural use.
2. *Compare speech samples within speakers.* To control for accent, proficiency, and other potential confounding inter-speaker variables, ratings should be elicited for the same speakers with and without PM use. To avoid any influence of raters' subsequent rating on the same speaker, we employed a matched-guise technique (used widely in sociolinguistics research to determine covert attitudes towards accents, dialects, and other linguistic features, Giles & Billings, 2004) with distractors between focal listening samples so that raters would fail to recognize that they were rating a repeated dyad in the survey.
3. *Employ principled (data-driven) PM use in the stimuli while prioritizing naturalness.* We followed Fung and Carter's (2007) taxonomy of PM types as a starting point but collected baseline data from eight L1 speakers to better understand authentic frequency and placement of PMs to guide the control of their use in the speech samples. Baseline data was obtained from L1 speakers because L2 speakers have been found to use PMs differently than L1 speakers in classroom contexts (Fung & Carter, 2007).
4. *Conceal the research focus of PMs from raters.* Considering that PMs have at times been discounted as features of disfluency (Trenchs-Parera, 2009), unclear thinking (Crystal, 1988), or bad habits (e.g., overuse of like), we did not want PMs to attract attention as the focus of the study to minimize potential influence from existing preconceptions. To support a covert focus on PMs, we limited participation to lay listener raters by excluding respondents who had studied or were studying linguistics/applied linguistics, or who self-identified as L2 teachers. We also opted to limit the description of fluency in the prompt (rather than attempt to operationalize interactive fluency in terms of the interactive/cognitive function of PMs) to help reveal lay listeners' subjective notions of fluency.

To achieve these goals, two surveys were designed with embedded speech samples, each focusing on a different dyad of advanced L2 speakers of English (four speakers altogether). Each survey included two distinct audio segments of about 75 s extracted from the same extended conversations. The use of PMs was controlled in both segments partly through digital editing and three distractor conversations from different dyads were included between the focal segments to prevent listeners from realizing they were rating the same speakers twice at the beginning and end of the survey. Participants were recruited to rate the fluency of each speaker in a series of short dyad conversations and quantitative ratings were followed by open-ended comments for justifications. We used mixed-effects modeling to predict fluency ratings from our target independent variables (PM condition, listener status, and their interaction) while controlling for the random effects of listener, speaker, and survey.

2.2. Participant raters

A total of 290 listeners were recruited via snowball sampling among contacts at universities in the U.S. and multiple international contexts to include a variety of L1/L2 backgrounds and L2 proficiency levels. Participants were provided informed consent to rate speech samples; however, 59 participants were excluded due to missing data, gibberish comments, or reported low L2 proficiency. L2 speaker respondents who listed their proficiency as low intermediate or below were excluded after evidence in piloting showed that these listeners frequently rated beginner L2 speakers (distractor dyads in the survey) higher than advanced L2 speakers, and in some cases, higher than L1 speakers (another distractor). In total, ratings from 231 listeners (180 L1 listeners and 51 L2 listeners) were included in the analysis. The L2 listener rater participants represented a variety of L1 backgrounds (e.g., Bosnian, Chinese, Dutch, Japanese, Lithuanian, Polish, Spanish, Thai, Vietnamese).

2.3. Speech samples

Six highly advanced L2 English graduate students studying applied linguistics were enlisted to record conversations in dyads for the matched-guise test. Male/female speakers were paired so listeners could distinguish between interlocutors for individual ratings. To control for topic, speakers were provided with several conversation prompts about their recent shared experience of taking a high-stakes test in their graduate program (e.g., strategies for preparation, feelings about the test) to elicit 20 min of authentic conversation. We anticipated the emotional topic would lead to more meaningful and empathetic conversational involvement with marking of shared knowledge and assessment of interlocutors' knowledge of the topic (i.e., eliciting greater use of cognitive and interpersonal functions of PMs). The recorded conversations were interrupted halfway through for the researcher to introduce and explain the functions of cognitive and interpersonal PMs and provide a list of them from Fung and Carter (2007), encouraging speakers to highlight PMs in each category they felt comfortable trying to use naturally in the final 10 min of the conversation. The second half of the

conversation was used to construct a speech sample with PM use to minimize the need for digital manipulation in their control. Despite their advanced proficiency, we anticipated the L2 speakers to lack natural control over each PM type that we hoped to employ in the stimuli. To ensure sufficiently diverse PM use we also recorded each speaker individually using each of the listed PMs in isolation, unscripted but within the context of larger utterances, to capture natural speed and alliteration for later insertion as needed to approximate expert-like use (described below).

To create the listening stimuli for comparison for each dyad, we identified clips (both individual utterances and sequences of turn-taking) from before and after the break in the recordings to recreate two edited segments, each of approximately 75 s with balanced participation between interlocutors. Different segments were used for comparison rather than repeating the same segments with the presence/absence of PMs to prevent raters from realizing they were hearing the same speakers twice (protecting the matched-guise). We then followed an extensive process to control for linguistic features (temporal and other) that are known to significantly correlate with perceived fluency ratings to prevent these potential confounding influences (see Appendix A for details of this process, accessible with data, materials, and analysis code via the Open Science Framework: https://osf.io/yb8dk/?view_only=a9fbb17304744668b54369781c47a76e).

Finally, we controlled for PM use in each stimuli first by digitally removing all naturally occurring cognitive and interactive PMs from the –PM segments. To guide principled deployment of PMs in the +PM segments we collected baseline data from four L1 speaker dyads (also graduate students studying in the same program) who performed the same conversational task. A 3-min segment in the middle of each L1-L1 conversation was transcribed to analyze the frequency (per 1000 words), variety, and placement of cognitive and interpersonal PM use comparing utterance-initial, -medial, -final, or in isolation as a backchannel (see Appendix B in the supplementary materials for these results). We controlled for the use of PMs in the +PM segments following patterns in the baseline data. For example, our baseline data revealed that PMs comprised between 9 and 19% of L1 interlocutors' speech (from word count), and so we kept within those parameters, loosely following the ratio of interpersonal and cognitive PMs found in the baseline data (70/30) while ensuring representation across the most common functional categories (Table 1 presents the types and frequencies of each PM utilized across speakers). Most PMs occurred naturally in the conversation but 36% of them were digitally inserted, predominantly in the utterance initial position, followed by mid-utterance and backchannel placement.

The resulting + PM segments do not perfectly balance the frequencies of all PM types across speakers because we deemed that as unnatural, especially in view of our baseline data that revealed considerable individual variation (in preference/style). And since not all the PMs we recorded in isolation sounded natural for all speakers, we prioritized authenticity and avoided forcing their use, which we suspected would also help avert attention to PMs as the focus of the study. It was crucial that speakers appeared to own their use of PMs (which was confirmed in piloting), and so not all PMs were conducive in this aim across speakers. More detailed description of how the +/-PM segments were manipulated and piloted to ensure they sounded natural is presented in Appendix A and sample transcripts of the edited + PM/-PM segments are supplied in Appendix C of the supplementary materials.

2.4. Survey design

Two versions of the survey (each with a different focal dyad) were created to limit the length to about 20 min, as early piloting revealed rater fatigue when both comparisons were included in one survey. Two versions were created for each of the two surveys, each with the same listening samples but with focal segments and some distractors presented in different orders (A and B) for a counterbalanced design. Table 2 presents the order of listening samples for each of the four survey versions and the pairing of speakers', their L1s, and general proficiency.

The distractor conversation samples were obtained from three other dyads, also discussing a similar testing experience. The

Table 1
Type and frequency of PMs used by each speaker.

Survey #, Speaker L1, PM frequency	Interpersonal			Cognitive	
	Marking shared knowledge	Indicating attitudes/stance	Showing response	Denoting thinking process	Elaboration
1, Male/Japanese, Int = 9 Cog = 5 Tot = 14	you know (1)	actually (1) exactly (1)	yeah (6)	I think (1) well (2)	like (1) I mean (1)
1, Female/Russian, Int = 12 Cog = 5 Tot = 17	you know (3)	actually (1) really (1) exactly (1)	yeah (5) really? (1)	well (3) okay (1)	I mean (1)
2, Male/Arabic, Int = 10 Cog = 4 Tot = 12	you know (1)	actually (1) definitely (1)	yeah (4) right (1)	I think (1) well (1)	like (1) I mean (1)
2, Female/Korean, Int = 9 Cog = 6 Tot = 15	you know (2)	just (1) sort of (1)	yeah (3) sure (1) okay (1)	I think (1) well (1)	I mean (4)

Table 2
Order of listening samples in the surveys.

Speakers' L1s	Survey 1: -PM Advanced L2 (Focal dyad 1)		Survey 2: +PM Advanced L2 (Focal dyad 2)	
	M: Japanese F: Russian		M: Arabic F: Korean	
Survey Order	A	B	A	B
1st sample	-PM Focal dyad 1	+PM Focal dyad 1	-PM Focal dyad 2	+PM Focal dyad 2
2nd sample	^a Distractor A	Distractor B	Distractor A	Distractor B
3rd sample	^b Distractor B	Distractor A	Distractor B	Distractor A
4th sample	Distractor C: Beginner proficiency L2 dyad (M: Chinese; F: Portuguese)			
5th sample (guised; repeated dyad from 1st sample, different conversation)	+PM Focal dyad 1	-PM Focal dyad 1	+PM Focal dyad 2	-PM Focal dyad 2

^a Distractor A: Advanced L2 speaker dyad, Female Spanish L1, Male Arabic L1.

^b Distractor B: L1 speaker dyad, American graduate students.

distractor dyads included another pair of advanced L2 English speakers, a dyad of L1 English speakers, and a low-proficiency L2 English speaker dyad to provoke a challenge in comprehension, which we anticipated would aid in distracting raters from the guise (Kang & Rubin, 2009). The low proficiency distractor was presented in the same position (4th listening sample) in all survey versions so that the same relatively disfluent conversation would have the same potential priming effect on the final focal segment in all versions.

Participants were instructed to listen to each recording and rate the fluency of each individual speaker (male and female separately) with a continuous sliding scale from 0 to 100, following Crowther et al. (2015). We avoided presenting a detailed descriptor of fluency in the scale (following Kormos & Dénes, 2004) for two reasons. First, we aimed to conceal the research focus of PMs because at times they have been discounted as features of disfluency (e.g., Trenchs-Parera, 2009). To avoid eliciting ratings influenced by such negative preconceptions, we avoided describing PMs or their functions as indicators of fluency (note that piloting revealed success in concealing the covert focus of PMs in the study, as none of the 22 raters in piloting mentioned PMs when asked their perception of the research focus after taking the survey).

Second, a goal of this study is to develop understanding of the components that contribute to lay perceptions of interactive fluency. Limiting the descriptor would enable listeners to operationalize interactive fluency subjectively (rather than be supplied a definition). We included a general description of varying degrees of “smooth speech” to guide listeners away from the notion of fluency as general proficiency, towards the narrow sense of the term (see Appendix D of the supplementary materials for the rater interface).

Qualitative data was collected to supplement the quantitative data in an explanatory mixed-methods design. Following the ratings for each of the five audio samples, an open-ended question required participants to justify their ratings for each speaker (following Kormos & Dénes, 2004; Rossiter, 2009). We did not anticipate that raters would explicitly point to PMs as they were purposefully concealed as a study focus. Rather, we were interested in the potential covert influence of PMs in shaping perceptions of temporal features of speech that are known to be associated with fluency (albeit in monologue), as we assume these features remain the dominant influence on the perception of interactive fluency.

2.5. Analysis

For the quantitative analysis, we used the *lme4* package (Bates et al., 2015) in R (Version 4.1.0; R Core Team, 2021) to conduct linear mixed-effects models to analyze the impact of PM condition and listener L1-L2 status on fluency ratings while controlling for the possible random effects of listener, speaker, and survey. Random intercept models were fit to determine the most appropriate random effect grouping before the fixed effects were included. Models were compared using AIC (Akaike Information Criterion) and the ANOVA function to determine whether additions improved the model fit significantly (Field et al., 2012). Alpha levels were set to $p < .05$ for the analyses, and R^2 effect sizes (both marginal and conditional) are reported and interpreted following Plonsky and Ghanbar's (2018) field specific recommendations: $R^2 < 0.18$ = small, $R^2 > 0.51$ = large. The R code and data used for this analysis are included in the supplementary materials. The internal consistency of each survey was acceptable ($\alpha = 0.75$ for survey 1 and $\alpha = 0.68$ for survey 2).

Raters' qualitative justifications were analyzed by coding for two dimensions in the data: *implicit* and *explicit*. If a rater explicitly referenced or alluded to PMs (with or without technical language identifying them as such, e.g. ‘filler words’), we compared their two corresponding justifications for the same speaker and coded as *explicit fluency* when PMs were described favorably (e.g., ‘she was able to express herself using an appropriate variety of expressions like, ‘yeah,’ ‘you know’ that L1 speakers usually use’) and as *explicit disfluency* when PM use was described unfavorably (e.g., ‘he used fillers such as “yeah” and “uhm” which made his sentences less fluent’).

In designing the survey to conceal the covert research focus of PMs, we expected that few raters would comment on PMs explicitly, failing to notice them or not knowing how to refer to them if they did. Consequently, we also analyzed the qualitative data to uncover the potential subconscious (*implicit*) influence of PMs on perceptions of the temporal aspects of fluency (speed, pausing phenomenon, repair, and flow), as we anticipate that temporal features remain the driving influence on perceptions. As we carefully controlled for temporal features between $\pm$ PM segments, we wanted to explore whether PMs could influence perceptions of them. For example, if a

Table 3
Examples of implicit coding.

Coding	Comment on + PM performance	Comment on -PM performance
Implicit fluency	The male had a good handle on the language and was relaxed.	I could hear pauses here and there when he would be thinking about how to say his next thought.
Implicit disfluency	The pronunciation of some vowels in words was noticeable, but he was understandable.	No hesitations, rapid speech.
Implicit Neutral	Speaks at a very normal speed with good pronunciation. She seemed to comfortably express herself. <i>Uses expressions like "you know?" like a native speaker.</i>	Pronunciation is very good and she speaks at a good pace .

rater described the speaker in the –PM segment as slower, choppy, or using more pauses relative to the same speakers' corresponding + PM segment (even though temporal features such as pace, speech rate, number and average length of pauses, pause location, repairs, etc., were all balanced in the design), we assume the biased discrepancy in perception can be attributed to PM use, the only notable difference in the stimuli. Table 3 presents examples of coding for *implicit fluency*, *implicit disfluency*, and *implicit neutral*.

In the example of *implicit fluency* 'pauses' are mentioned only in reference to the –PM condition, while in the *neutral* example, temporal aspects ('speed' and 'pace') are described as positive attributes in both performances. The *neutral* example is also an example of coding for *explicit fluency* with explicit reference to favorable use of a PM ('you know'). In the implicit coding we excluded features not typically associated with fluency in its narrow conceptualization (e.g., vocabulary choice, accuracy, accent, sentence structure—see Appendix E for the complete coding scheme). Two authors coded 15% of the comments separately reaching an initial inter-rater agreement of 76%. After discussing discrepancies, we continued to code an additional 15% of the data to result in an acceptable agreement of 87%. Any further coding that caused doubt was discussed to reach agreement.

3. Results

Respondents were directed to one of four different versions of the survey: two with different dyads in the targeted conversations (surveys 1 and 2) and two versions of each of those with a different order of presentation of the targeted conversations (A and B) (see Table 4).

The mean fluency ratings (using a 0–100 scale) for + PM and –PM segments and standard deviations are displayed separately for each of the four speakers in Table 5. Descriptive statistics show that ratings for all four speakers were relatively higher for the +PM segments with mean differences in ratings of 4.34, 10.30, 5.34 and 4.10. L1 listeners consistently rated + PM segments higher in fluency than their –PM counterpart performances; L2 listeners showed more variability, with a preference similar to the L1 listeners for + PM segments for the L1 Russian speaker, but little difference between + PM and –PM segments for the other three speakers.

Figs. 1 and 2 present violin and boxplots of ratings of each speaker in each condition (+PM, –PM), separating L1 and L2 listener rater data, visually distinguishing a slight advantage in ratings for + PM over –PM segments, particularly for the L1 raters. The figures also show the highest fluency ratings overall were given by both L1 and L2 listener raters to the Korean L1 speaker.

To answer the research questions, we used mixed-effects modeling to predict fluency ratings from the target independent variables (PM condition, listener status, and their interaction) while controlling for the random effects of listener, speaker, and survey. First, random intercept models were fit to determine the most appropriate random effect grouping, starting with listener, then speaker, and then survey. Including by-speaker random intercepts significantly improved model fit, $\chi^2(1) = 182.73, p < .001$, but the inclusion of by-survey random intercepts resulted in a model that failed to converge. Therefore, the by-listener and by-speaker random effects were retained in the model. Model assumptions were checked via visual inspection of plots: Linearity was tested by plotting the model residuals to the outcome variable, homogeneity of variances was tested by plotting the model residuals to the predicted values, and normality for model residuals was tested with a QQplot (see Appendix F). Table 6 provides a summary of the model.

As seen in Table 6, there was a significant negative effect for PM condition such that.

–PM samples were rated lower than + PM samples. The fixed effect estimate for PM condition indicates that, on average, –PM samples were rated approximately seven points lower than + PM samples on the 100-point scale. However, there was also a significant interaction effect such that L1 listeners rated –PM segments lower than + PM segments but L2 listeners did not. This can be seen in Fig. 3, which plots the interaction effects. As indicated by the marginal $R^2 = 0.03$, the fixed effects only explained 3% of the variance in fluency ratings which can be interpreted as a small to negligible effect (Plonsky & Ghanbar, 2018).

Results of the coding of the qualitative comments that followed the numeric ratings are presented in Table 7, divided into implicit and explicit frequency counts and percentages for each speaker. Implicit counts represent instances when rater's qualitative justifications associated with fluency were unbalanced in favor of +PM segments (+ for *fluency*), in favor of –PM segments (– for *disfluency*), or neutral (Neut). Explicit coding represents comments with direct reference to PMs as descriptors of either fluency (+) or disfluency (–).

Qualitative results support the quantitative findings in that raters described speakers more favorably in terms of temporal fluency when they used PMs than when they did not, even though temporal features were balanced in the design. L1 listeners ascribed relatively greater temporal fluency to + PM segments (32.8% as fluency) nearly twice as often as they did to –PM segments (16.7% disfluency), compared to 31.4% fluency and 19.6% disfluency by L2 listeners, respectively. Furthermore, explicit mention of PMs only

Table 4
Survey respondents.

Survey # (Dyad)	Survey Version (Order)	L1 Speaker Respondents	L2 Speaker Respondents	Total Respondents
1	A	36	13	49
	B	40	8	48
	Total	76	21	97
2	A	55	10	65
	B	49	20	69
	Total	104	30	134
Surveys combined		180	51	231

Table 5

Fluency ratings between segments with and without pragmatic markers.

Survey	Speaker	Survey Version (Order)	L1 Speaker Respondents		L2 Speaker Respondents		Total Respondents	
			PM + M (SD)	PM–M (SD)	PM + M (SD)	PM–M (SD)	PM + M (SD)	PM–M (SD)
1	Male, Japanese L1	A	79 (13.1)	67 (19.2)	72 (25.2)	69 (16.0)	77 (17.2)	68 (18.3)
		B	69 (19.1)	70 (18.7)	70 (15.0)	72 (12.8)	69 (18.3)	70 (17.8)
		Total	74 (17.2)	69 (18.9)	71 (21.4)	71 (14.6)	73 (18.1)	69 (18.0)
	Female, Russian L1	A	82 (12.6)	66 (19.2)	83 (19.8)	71 (13.7)	82 (14.6)	67 (17.9)
		B	74 (16.3)	69 (17.3)	82 (11.5)	75 (10.6)	75 (15.8)	70 (16.4)
		Total	78 (15.1)	68 (18.2)	83 (16.8)	72 (12.5)	79 (15.5)	69 (17.2)
2	Male, Arabic L1	A	76 (16.5)	61 (20.9)	72 (15.9)	60 (12.7)	76 (16.4)	61 (19.8)
		B	70 (18.1)	71 (15.9)	60 (19.4)	69 (14.1)	67 (18.9)	70 (15.3)
		Total	73 (17.5)	66 (19.2)	64 (18.9)	66 (14.1)	71 (18.2)	66 (18.2)
	Female, Korean L1	A	90 (10.7)	81 (17.2)	90 (14.5)	90 (9.9)	90 (11.2)	83 (16.6)
		B	85 (14.3)	83 (12.7)	86 (17.1)	88 (11.9)	85 (15.1)	85 (12.6)
		Total	88 (12.8)	82 (15.2)	87 (16.1)	89 (11.2)	88 (13.5)	84 (14.6)

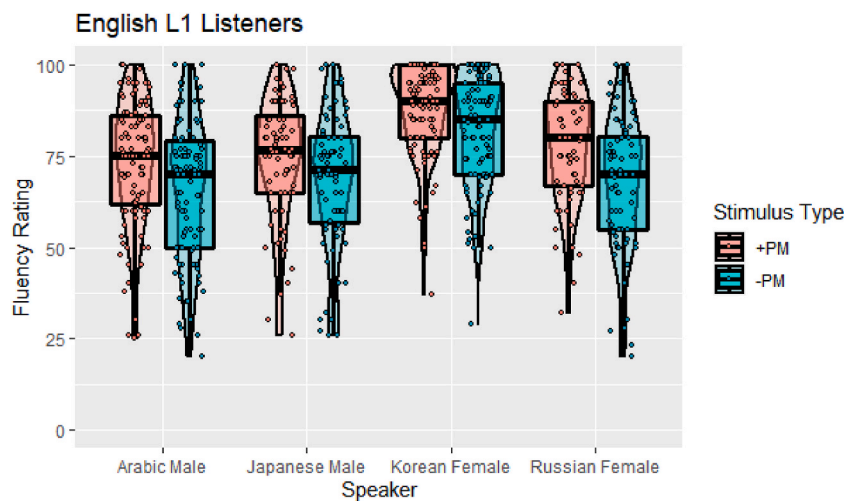
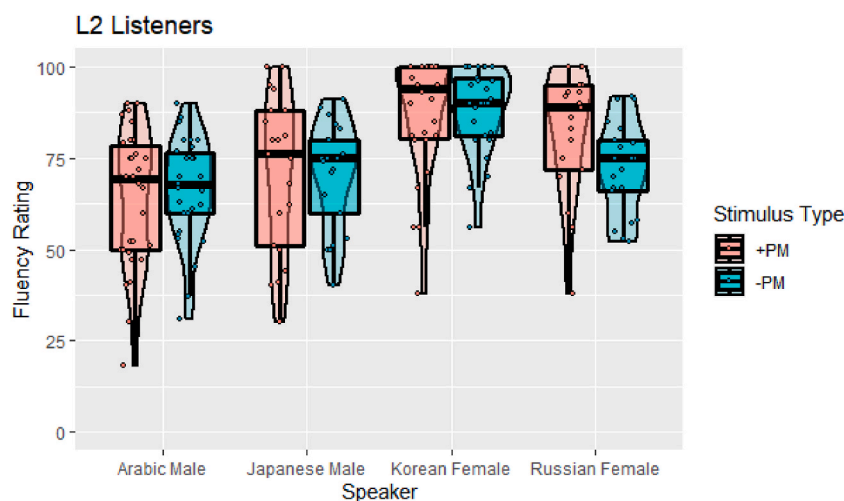
**Fig. 1.** Violin and boxplots of the ratings in each condition (+PM, –PM) for the L1 raters separated by speaker.**Fig. 2.** Violin and boxplots of the ratings in each condition (+PM, –PM) for the L2 listener raters separated by speaker.

Table 6
Summary of mixed-effects model fit to fluency ratings.

Fixed effects	<i>b</i>	<i>SE</i>	95% CI	<i>t</i>	<i>p</i>
Intercept	78.23	3.93	[70.52, 85.94]	19.92	<.001
PM Condition	−7.16	0.97	[−9.06, −5.25]	−7.36	<.001
Listener Status	−2.57	2.57	[−6.89, −1.76]	−1.17	.244
Interaction	6.08	2.07	[2.02, 10.14]	2.94	<.001
Random effects	<i>SD</i>				
By-listener intercepts	10.39				
By-speaker intercepts	7.57				
Marginal R^2 /Conditional R^2	0.029/0.508				

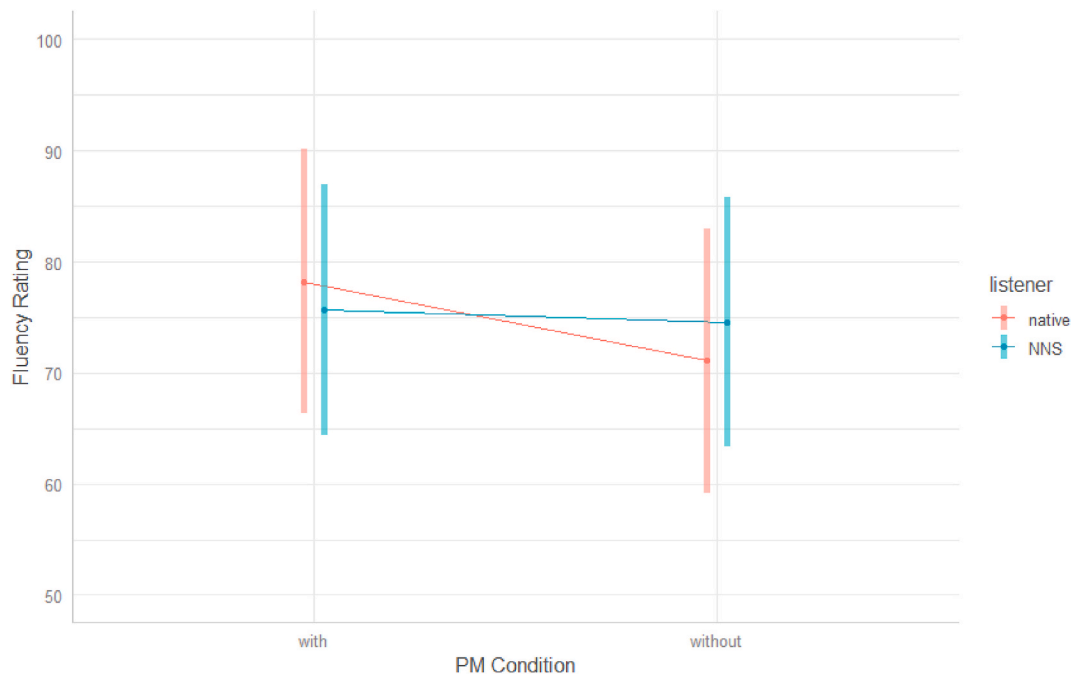


Fig. 3. Estimated marginal means and corresponding 95% confidence intervals for the interaction effect of the final model.

Table 7
Results from coding of rater comments.

Survey	L1 Listener Raters (Survey 1, <i>n</i> = 76; Survey 2, <i>n</i> = 104)					L2 Listener Raters (Survey 1, <i>n</i> = 21; Survey 2, <i>n</i> = 30)				
	<u>Implicit</u>			<u>Explicit</u>		<u>Implicit</u>			<u>Explicit</u>	
	Neut.	+	−	+	−	Neut.	+	−	+	−
1 M	37	26	13	1	2	11	6	4	2	0
%	48.7	34.2	17.1	1.3	2.6	52.4	28.6	19.0	9.5	0
1 F	37	30	9	1	1	7	11	3	1	2
%	48.7	39.5	11.8	1.3	1.3	33.3	52.4	14.3	4.8	9.5
2 M	49	32	23	1	1	14	8	8	0	0
%	47.1	30.8	22.1	1	1	46.7	26.7	26.7	0	0
2 F	59	30	15	8	6	18	7	5	2	3
%	56.7	28.8	14.4	7.7	5.8	60.0	23.3	16.7	6.7	10
Total	182	118	60	11	10	48	32	20	5	5
%	50.6	32.8	16.7	3.1	2.8	47.1	31.4	19.6	4.9	4.9

occurred in about 3–5% of raters' comments (e.g., 'However, he used air fillers for example, *like, you know, I mean*. The fluency was ok regardless'). L1 and L2 listener raters described PMs both favorably (3.1% and 4.9%, respectively) and unfavorably (2.8% and 4.9%), as in the prior example, to a similar extent.

4. Discussion

We set out to determine if the use of cognitive and interpersonal PMs (those found to be infrequently used in L2 discourse in prior research) are perceived as favorable in spoken dialogic fluency. Our quantitative results suggest that raters generally perceived the use of these PMs by L2 speakers as slightly favorable; nevertheless, the magnitude of the effect was quite small such that PM condition only explained 3% of the variability in fluency ratings. On the one hand, we might not expect PM use to have large explanatory power on predicting fluency ratings given the complex set of linguistic features that have been shown to contribute to ratings of perceived fluency (e.g., utterance fluency characteristics such as speech rate and pausing, Bosker, Pinget, Quene, Sanders, & De Jong, 2013; morphological and phonological accuracy, Suzuki & Kormos, 2020). In other words, PM use is just one of many potential features in the speech stream that a listener might rely on when judging fluency. That being said, the current study provides evidence that PM use has a statistically significant impact on ratings of perceived fluency albeit small in practical significance. This finding comes with an important caveat related to research question 2: the +PM advantage appears to be driven by L1 English listeners. The fact that L1 listeners were more sensitive to PM use than L2 listeners might not be surprising, especially given the range of proficiency levels of L2 listeners in the current study, and the fact that most of the L2 listeners in the sample took the survey in EFL contexts. In contexts where English is used as an international language (EIL), L2 English speakers likely interact predominantly with other EIL speakers and presumably have less exposure to such PMs in interaction or expectations for their use. Future research that controls for listener proficiency and English home language context could help confirm this finding. Furthermore, while the speakers in this study passed minimum TOEFL requirements for graduate school entrance, actual oral proficiency (e.g., OPI scores) could help in determining the role of speaker proficiency.

The qualitative findings provide exploratory insight into the underlying influence of PMs on perceptions of fluency, although they should be viewed cautiously in absence of statistical analysis. Raters rarely mentioned PMs in their justifications (less than 5%), which was expected in our effort to conceal the focus of the study. When they were mentioned explicitly, both L1 and L2 listeners commented favorably and unfavorably at the same rate. This was also unsurprising given that PMs have been viewed as disfluency phenomena, e.g., used to buy time in coping with breakdown, or as bad habits (Trenchs-Parera, 2009). In the implicit results, nearly twice as many participants described + PM speech as relatively more fluent in terms of temporal features (faster, with fewer pauses, and more flow) than the -PM segment, despite control over these temporal features in the design. More research is needed to confirm the apparent covert influence of PMs on fluency perceptions, but a possible explanation for these results could be that temporal features share the dominant influence on fluency perception in both monologic and dialogic performances, and that the peripheral role that PMs play in interactive fluency is less noticeable, likely overshadowed by relatable (i.e., surface level) temporal indicators when listeners are pressed to analyze their ratings.

An interesting finding peripheral to the focus of this study, but that supports the notion that PMs can at times be discounted as features of disfluency, was the apparent bias we noticed in raters' qualitative descriptions of PM use between L1 and L2 speakers. Although this study focused on L2 speakers' use of PMs, the L1 speaker dyad included in the survey as a distractor presented data for comparison. We discovered that of the 16 respondents (6.9% of the sample) who explicitly described PM use as disfluency, only four of them (three L1 listeners and one L2 listener) did so for both L2 speakers and the L1 speakers equally, while 11 (seven L1 and four L2 listeners) commented negatively about PM use for L2 speakers but interestingly made no mention of PM use by either speaker in the L1 speaker dyad, despite the comparable frequency, variety, and placement of PMs in the L2 speaker segments. Alternatively, only one respondent (L1) described L1 use of PMs as disfluency while not commenting about PM use for the L2 speakers. This appears to demonstrate potential bias in perceptions of the use of PMs disfavoring L2 speakers, whereas perceived accent or L2 status likely signals doubt in the judgment of the appropriateness of PM use. Although PM use appears to result in perceptions of higher fluency for advanced L2 speakers, relative to their ratings of L1 speakers, some listeners appear to (unjustly) assume L2 speakers are using PMs as a crutch to hide their disfluency. More research is needed to confirm this speculated bias, and a study design that also includes speech samples by L1 speakers without PM use could help. Furthermore, corresponding experiments in non-English home language countries where perceptions of PMs and of perceived accent may be different could help us understand the potential moderating contributions to fluency perceptions.

With regard to pedagogical implications, we recognize that amid admirable effort in the field of L2 teaching to abandon "native speaker" norms in favor of goals linked to intelligibility (e.g. growing recognition of World Englishes), L2 practitioners may feel reluctant to prioritize PMs in speaking curricula—features that, to some extent, are also linked to differences in individual style. It is important to acknowledge, however, the role PMs play in developing and maintaining a fluent interaction. Most educational practices and activities that have been reported to have a positive effect on the development of L2 cognitive fluency focus on the development of automaticity and involve pre-task planning time, task repetition, and the 4/3/2 technique (Tavakoli & Hunter, 2018). Whereas automaticity plays a central role in smooth performance, these activities generally feature fluency practice in monologue. A supplemental instructional focus on the functions of interactional features that promote fluid dialogue could serve a crucial role in a balanced fluency training curriculum.

We argue for a language awareness approach to the teaching of PMs to enhance learners' perceived interactive fluency with an emphasis on learners' ability to co-construct fluency at the interactional level. The type of pedagogical trajectory we envision would involve the analysis of data (i.e., a transcript), explicit teaching of PM functions and positioning in the utterance, close-ended

functional transcript analysis, followed by open-ended data (e.g., corpus) navigation for the potential identification of other PMs with similar functions, repeated oral practice through language tasks, and free recorded practice with a fluency rubric (see Hasselgren, 2002 for sample descriptors). Developing activities for the teaching of PMs that focus on raising learners' language awareness of their multiple functions in dialogue would go a long way in providing learners with a tangible 'fluency enhancement strategy' (Götz, 2013, p. 41) – they could incorporate PMs into their speech, and consequently appear more fluent in naturalistic interaction, if they so choose. As a corollary, the learners would also be equipped with receptive knowledge that would contribute to their navigation of functional pragmatic nuances in decoding the implicit illocutionary intent behind the use of PMs, and in doing so hone their L2 listening skills.

5. Conclusions

Perceptions of fluency (or lack thereof) have social consequences, requiring a deeper understanding of the aspects of speech that contribute to perceived fluency, particularly in interaction. These results provide empirical support for the perceived value of cognitive and interactive PMs in dialogue, as raters generally judged speakers favorably when they employed expert-like use of them in interaction. We argue that these results suggest the need for increased pedagogical attention to these less commonly taught PM types, as practitioners should carefully consider their role in developing competence in spoken interaction.

This study motivates further investigation into how and when these PM types can serve L2 learners in becoming more fluent speakers in interaction, which we presume to be highly dependent on contexts and purposes for language learning. Further research is needed to investigate how PM type and their specific functions contribute to fluency, along with the interaction between PM use and other phenomena associated with automaticity and accommodation strategies in discourse (e.g., turn construction). Better understanding of the impact of PMs on perceived fluency can also help guide reliable and valid operationalization of fluency in dialogue, as monologic measures are insufficient to capture the components of successful co-construction of meaning in interaction.

Credit statement

Dan Brown: Conceptualization, Methodology, Validation, Formal Analysis, Investigation, Writing – Original Draft, Writing – Review & Editing, Project administration. Julieta Fernández: Conceptualization, Methodology, Formal Analysis, Writing – Original Draft, Writing – Reviewing & Editing, Supervision. Amanda Huensch: Formal Analysis, Data Curation, Writing – Review & Editing, Visualization.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.system.2023.103157>.

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